Census and Urbanisation Topic 2 — Population Growth, Demographic Transition and Theories

Current Affairs

Year	Fact	Why it matters
Rolling	Refresh edition-bound figures from latest Census / Survey / State CA desks.	Do not invent interim totals.

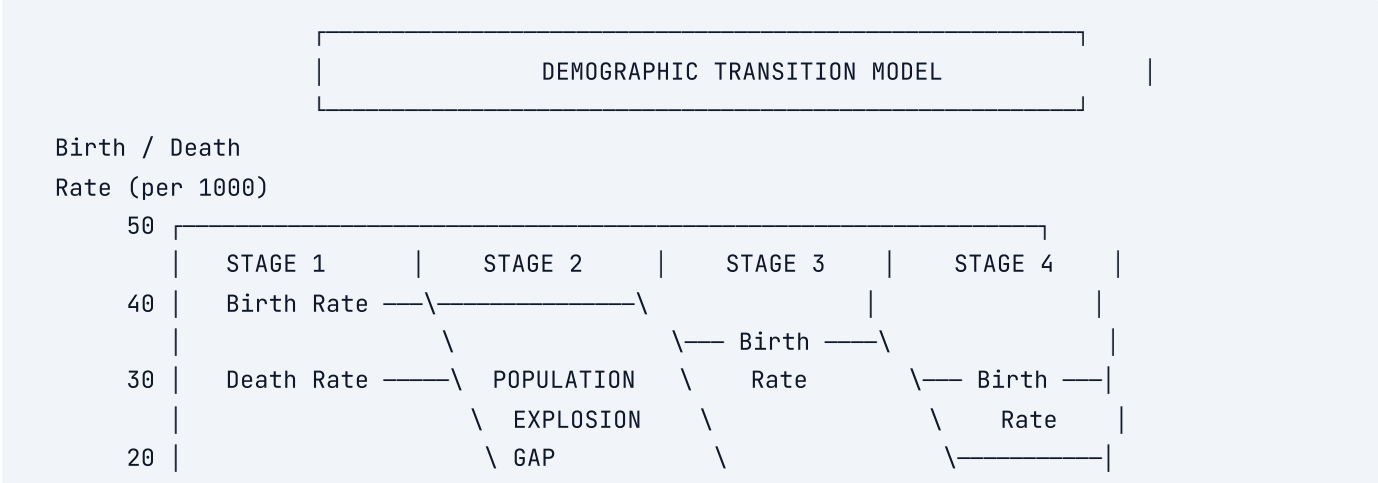
UP district extremes also → UP Special Topic 6. This chapter owns the national Census / demography desk.

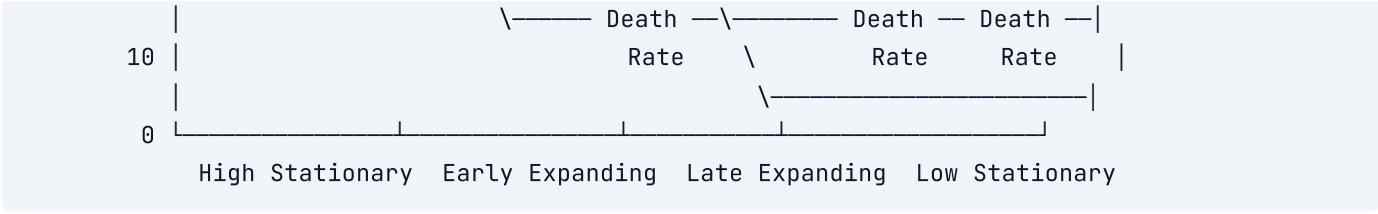
Teaching cards

2.1 Demographic Measures & Growth Mechanics

A. Basic Demographic Rates & Formulas

1. **Natural Growth Rate of Population**: ext{Natural Growth Rate} = ext{Crude Birth Rate (CBR)} - ext{Crude Death Rate (CDR)} *(Note: Does not account for international migration. Overall Population Growth = Natural Growth + Net Migration).*
2. **Decadal Growth Rate**: ext{Decadal Growth Rate (\%)} = \left(rac{P_t - P_0}{P_0} ight) imes 100 Where P_t = ext{Population at current census} and P_0 = ext{Population at preceding census 10 years ago}.
3. **Annual Exponential Growth Rate (r)**: Calculated using the compound continuous growth formula P_t = P_0 \cdot e^{rt}. - India's annual exponential growth rate dropped from **1.97%** (1991–2001) to **1.64%** (2001–2011).
4. **Population Doubling Time (Rule of 70)**: ext{Doubling Time (Years)} pprox rac{70}{ ext{Annual Growth Rate (\%)}} *(Example: If growth rate is 1.64%, doubling time is 70 / 1.64 pprox 42.7 ext{ years}).*





2.2 In-Depth Critique of the Malthusian Model

A. Why Malthus Failed to Foresee 19th & 20th Century Reality

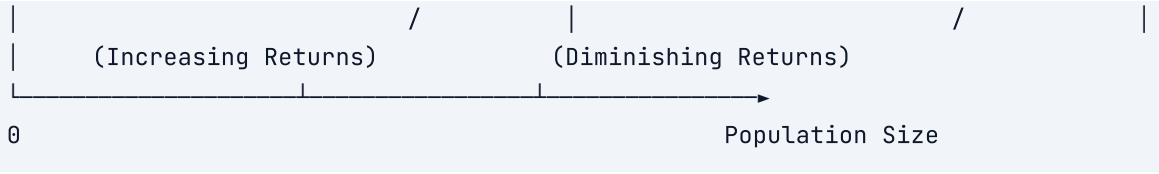
Thomas Malthus published his pessimistic treatise in 1798 on the eve of the Industrial Revolution. While his mathematical formulation became a classical benchmark, his predictions failed in Western Europe and the Americas due to fundamental technological and social oversights: 1. **The Agricultural Revolution & Technological Innovations**: - Malthus failed to foresee mechanisation (tractors, combine harvesters), chemical fertilizers (Haber-Bosch nitrogen fixation), synthetic pesticides, tube-well irrigation, and biotechnology (Green Revolution / HYV hybrid seeds) which expanded food production exponentially rather than arithmetically. 2. **Transportation & Global Food Supply Chains**: - Development of steamships, railways, and refrigerated maritime shipping enabled mass transport of grain from agricultural frontiers (Americas, Australia, Ukraine) to densely populated European industrial centres, averting regional famines. 3. **Voluntary Birth Control & Rising Living Standards**: - Malthus believed humans would perpetually reproduce up to the subsistence limit. He failed to anticipate the demographic transition, where rising education, urbanisation, female workforce participation, and modern contraception caused birth rates to decline voluntarily with rising per capita income. 4. **Distinction between Biological Capability and Human Workforce**: - Malthus viewed a newborn baby solely as "another mouth to feed", ignoring that each person is also equipped with "two hands to work" and a brain to innovate capital and technology.

2.3 Optimum Population Theory & Dalton's Formulation

A. The Mechanics of Optimum Population

Developed by British economist **Edwin Cannan** (1924) in *Wealth*, the Optimum Population Theory moves away from Malthusian biological subsistence toward **per capita economic productivity and living standards**. - **Underpopulation**: When the population of a country is too small relative to its natural resources and capital stock. In this state, the full benefits of division of labour, economies of scale, and infrastructure utilization cannot be realized. An increase in population leads to **increasing returns and rising per capita income**. - **Optimum Population Point (O)**: That precise population size where the Law of Diminishing Returns is exactly balanced by the economies of large-scale production, resulting in the **highest possible output per head**. - **Overpopulation**: When population exceeds the optimum point. The pressure of population on fixed natural resources and capital causes the Law of Diminishing Marginal Returns to dominate, resulting in **declining per capita income and falling living standards**.

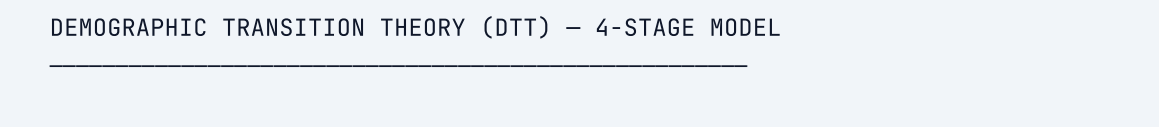




Bilingual Terminology — Population Growth and Demographic Transition (जनसंख्या वृद्धि एवं जनांकिकीय संक्रमण)

English Term	हिंदी पारिभाषिक शब्द	Key Anchor
Demographic Transition Theory	जनांकिकीय संक्रमण सिद्धांत	W.S. Thompson (1929), Frank Notestein (1945)
Crude Birth Rate (CBR)	अपरिष्कृत जन्म दर	Births per 1,000 mid-year population
Crude Death Rate (CDR)	अपरिष्कृत मृत्यु दर	Deaths per 1,000 mid-year population
Total Fertility Rate (TFR)	कुल प्रजनन दर	Avg children per woman over reproductive life
Infant Mortality Rate (IMR)	शिशु मृत्यु दर	Deaths under 1 year per 1,000 live births
Maternal Mortality Rate (MMR)	मातृ मृत्यु अनुपात	Maternal deaths per 1,00,000 live births
Natural Increase	स्वाभाविक वृद्धि	CBR minus CDR
Replacement Level Fertility	प्रतिस्थापन स्तरीय प्रजनन	TFR ~2.1 for population stabilization
Demographic Dividend	जनांकिकीय लाभांश	Economic benefit of rising working-age proportion
Dependency Ratio	निर्भरता अनुपात	(0–14 + 65+) / (15–64) × 100
Population Momentum	जनसंख्या गति / संवेग	Growth continues even after TFR falls to replacement level
Malthusian Theory	माल्थस का जनसंख्या सिद्धांत	Population grows geometrically; food arithmetically
Optimum Population Theory	इष्टतम जनसंख्या सिद्धांत	Population level maximizing per capita output
Life Expectancy at Birth	जन्म के समय जीवन प्रत्याशा	Average years a newborn is expected to live

Extended Theory — Demographic Transition Stages (जनांकिकीय संक्रमण की अवस्थाएँ)



```
STAGE 1 — PRE-INDUSTRIAL (पूर्व-औद्योगिक)
┌──────────────────────────────────────┐
│  High CBR (40+) + High CDR (40+)     │
│  → Very slow / near-zero natural growth  │
│  → Pre-modern agrarian societies     │
└──────────────────────────────────────┘

STAGE 2 — EARLY INDUSTRIAL (प्रारंभिक औद्योगिक)
┌──────────────────────────────────────┐
│  High CBR (40+) + Declining CDR (20-30) │
│  → RAPID population growth           │
│  → Improvements in sanitation, medicine  │
│  India's Phase: ~1921-1951           │
└──────────────────────────────────────┘

STAGE 3 — LATE INDUSTRIAL (परिपक्व औद्योगिक)
┌──────────────────────────────────────┐
│  Declining CBR (20-30) + Low CDR (10-15)│
│  → Slowing population growth         │
│  → Urbanisation, education, family plan  │
│  India's Phase: ~1981-present (partly)   │
└──────────────────────────────────────┘

STAGE 4 — POST-INDUSTRIAL (उत्तर-औद्योगिक)
┌──────────────────────────────────────┐
│  Low CBR (~15) + Low CDR (~10-12)    │
│  → Very slow / near-zero / negative growth│
│  → Most of Europe, Japan             │
└──────────────────────────────────────┘
```

Population Theories — Comparative Overview

Theory / Theorist	Core Idea	Criticism	India Relevance
Malthus (1798) — *Essay on Population*	Population grows **geometrically** (2,4,8…); food grows **arithmetically** (2,4,6…). Positive checks (war, famine, disease) and preventive checks (celibacy, late marriage) regulate population.	Does not account for technological advances in food production; ignores demographic transition.	Malthusian fears were valid in pre-green-revolution India; less applicable post-1965.
Optimum Population Theory (Cannan, Carr-Saunders)	An **optimum population** exists for each resource level, maximizing per-capita output. Under-population and over-population both reduce output.	"Optimum" is dynamic and impossible to measure precisely.	Concept used in National Population Policy formulation.

Theory / Theorist	Core Idea	Criticism	India Relevance
Demographic Transition Theory (Thompson/Notestein)	Societies pass through **4 stages** of birth/death rate changes as they develop economically.	Cannot predict timing; assumes uniform development pathways across all societies.	India is transitioning from Stage 2 to Stage 3; some States (Kerala, TN) already at Stage 3.
Theory of Demographic Dividend	When working-age (15–64) population rises relative to dependents, savings, investment, and growth can accelerate	Dividend requires investment in health, education and job creation to be realized	India's demographic dividend window: ~2020–2050. Requires skill development.

One-Liner Revision — Population Growth and Demographic Transition

#	Fast Fact
1	DTT proposed by **W.S. Thompson (1929)** and elaborated by **Frank Notestein (1945)**
2	Stage 2 of DTT: High CBR + **Declining CDR** = rapid growth
3	India's peak decadal growth = **1961–71 (24.80%)**
4	Year of Great Divide = **1921** (–0.31% growth due to Spanish flu 1918)
5	India's CBR as per SRS 2020 ≈ **19.5** per 1,000
6	India's CDR as per SRS 2020 ≈ **6.0** per 1,000
7	Replacement level TFR = **2.1** children per woman
8	India's TFR (SRS 2020) ≈ **2.0** (near replacement level)
9	UP TFR ≈ **2.7** (above national average)
10	Demographic dividend window for India ≈ **2020–2050**
11	Dependency ratio India ~2011 = ~55 dependents per 100 workers
12	Malthus: population = **geometric**; food = **arithmetic**
13	Positive checks (Malthus) = **war, famine, disease**
14	Preventive checks (Malthus) = **late marriage, celibacy**
15	IMR India (SRS 2020) ≈ **28** per 1,000 live births
16	MMR India ≈ **103** per 1,00,000 live births (SRS 2017–19)
17	Life expectancy at birth India ≈ **69.7 years** (2015–19)
18	National Population Policy, 2000 — target TFR 2.1 by 2010 (not achieved)
19	Population Momentum = growth continues even after TFR falls to 2.1

#	Fast Fact
20	India became world's most populous country ≈ **April 2023** (surpassing China)